

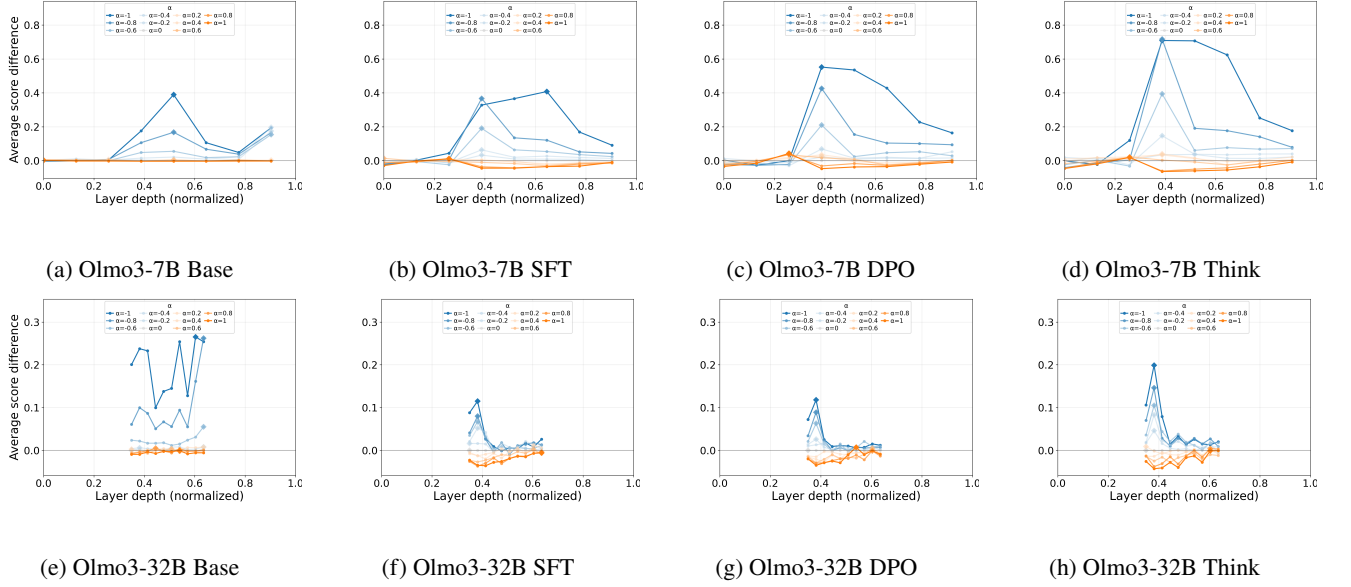


Figure 6: Steering response across training stages for the 7B and 32B model families. **(a–d)** For Olmo3-7B, steering becomes progressively more pronounced with training: peak gains (diamonds) increase, while troughs at suboptimal layers deepen, indicating greater sensitivity to internal representation shifts in later alignment stages. $l \in \{0, 4, \dots, 28\}$. **(e–h)** For Olmo3-32B, the response follows a U-shaped pattern: the Base model exhibits the strongest gains, which diminish after initial alignment; SFT and DPO remain largely unchanged, suggesting stabilization during preference optimization; and the Think stage produces a resurgence with stronger, more spatially localized peaks and troughs. $l \in \{22, 24, \dots, 40\}$. Even the strongest peak in the Olmo3-32B models (Base) is lower than the weakest peak in the Olmo3-7B models (also Base), underscoring the markedly reduced steering range of the larger model family.

to detect). Conversely, low verbalization rates do not imply the absence of an internal representation. Measuring only one facet of evaluation awareness may provide an incomplete picture of how a model responds to evaluation settings. These findings are relevant to broader discussions of strategic adaptation and behavior under oversight, although the results presented here do not by themselves establish such behaviors.

Limitations. The conclusions should be interpreted in light of certain limitations. First, we study a specific operationalization of evaluation awareness based on the distinction between benchmark prompts (GSM8K, TruthfulQA, and MASK) and WildChat prompts. Second, these prompt distributions also differ in topic, style, and length. However, since the probe is constructed on contrastive pairs that hold style and content fixed while swapping only evaluation- and deployment-indicative terms, the probe direction is unlikely to be dominated by surface-level dataset features. Third, verbalization is measured using a single LLM judge (GPT-5.4). Fourth, the causal analysis considers a single linear direction at one layer at a time; more complex interventions may produce different effects.

Future work. Several directions remain open such as non-linear probes, multi-layer and multi-direction interventions, alternative judge models, and evaluation settings that more closely resemble deployment environments. Understanding how evaluation awareness relates to broader forms of strategic adaptation also remains an important question.

6 Related work

Evaluation awareness. The term was introduced by [Needham et al. \(2025\)](#), who showed that frontier models can distinguish evaluation transcripts from deployment-like interactions. The phenomenon has since been observed in open-weight models ([Chaudhary et al. 2025](#); [Abdelnabi and Salem 2025](#); [Xiong et al. 2025](#)) and in proprietary models alike [Anthropic 2026](#), pp. 143–149; [Google 2025](#), pp. 24–25; [OpenAI 2026](#), p. 38), culminating in the International AI Safety Report, which lists it as a key development of the past year ([Bengio et al. 2026](#), pp. 10, 76). Some practical consequences are already visible: in one notable case, a third-party evaluator declined to issue a (mis)alignment assessment of a frontier model on the grounds that the rate of verbalized evaluation awareness made their evaluations unreliable ([Anthropic 2026](#), p. 118). [Chaudhary et al. \(2025\)](#) additionally report that the capability scales with model size across open-weight families, and [Devbunova \(2026\)](#) caution that probe-based evidence can be confounded by prompt format and structural regularities, calling for multiple complementary measurements rather than any single proxy. Accordingly, we provide a systematic study across multiple metrics, models, and training stages.

Adjacent alignment concerns. Evaluation awareness is situated within a set of concerns about systems that behave differently under oversight than in usual use, including situational awareness ([Berglund et al. 2023](#)), deceptive alignment ([Hobbhahn 2025b](#)), sleeper agents ([Hubinger et al. 2024](#)), and

alignment faking (Ngo, Chan, and Mindermann 2022).

Probing and steering. Linear probes have long been used to extract latent features from neural-network representations (Alain and Bengio 2016; Belinkov 2022; Turner et al. 2023). In the specific case of evaluation awareness, Hua et al. (2025) report that intervening on a model’s activations can attenuate evaluation-aware behavior, and shift outputs toward deployment-like. See Appendix J for more references.

7 Conclusion

We have presented the first systematic study of evaluation awareness across its representational, verbalized, and causal facets, measured on six open-weight models and tracked across the Olmo3 Base/SFT/DPO/Think checkpoints. Our findings show that evaluation awareness is linearly separable from the residual streams of every model we examine at some layers, only partially expressed in their outputs, somewhat controllable through linear interventions on the residual stream, and progressively sharpened by post-training. Interestingly, both probing and steering rely on a probe constructed from simple contrastive pairs, yet it generalizes to unseen benchmarks. These results argue for treating evaluation awareness as an empirical capacity of language models that cannot be ignored nor understood as a single metric.

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